Automatic ghostwriting of song lyrics

There are different approaches when it comes to automatic generation of song lyrics [1]. In most cases, song lyrics generation is conditioned on some musical constraint, such as melody and rhyme scheme. Another approach is ghostwriting [2]. Ghostwriting is the process of writing lyrics in the style of a specific artist. The purpose of the project is to build a system that generates song lyrics conditioned on an artist's style, focussing on traits like theme, phrasing, rhyme scheme, average length of lyrics or stanzas etc. A potential source of lyrics can be the Song Interpretation Dataset.

Prerequisites:
- Interest in Natural Language Processing and Machine Learning.
- Good programming skills (Python)
- Taking the module COM6115: Text Processing is highly recommended.

References:
[1] https://aclanthology.org/2020.nlp4musa-1.2.pdf
[2] https://aclanthology.org/W18-1604.pdf

(The project may require ethics approval, depending on the selected dataset and evaluation method.)

Lyrics-based music recommendation

There are two main approaches in music recommendation: collaborative and content-based, with collaborative being the most prevalent in music recommendation applications. Content-based recommendation doesn't focus on the listening habits of others, but rather analyses information extracted from song metadata, audio signals and song lyrics, with song lyrics being the most under-explored source. The purpose of this project is to build a system that uses song lyrics ~~(and potentially some audio-based features from sources like Spotify API)~~ for music recommendation.

Prerequisites:
- Interest in Natural Language Processing and Machine Learning.
- Good programming skills (Python)
- Taking the module COM6115: Text Processing is highly recommended.

Recommended reading:
- https://dl.acm.org/doi/pdf/10.1145/3405962.3405963?casa_token=gwTeRv3BNQ0AAAAA:4dFlwHLqDTTHSjU-uiZVlwkR7roYmyHiXGjc3NnaTwEdfnSOpaKlgrlRObM2iydUbWBfR_xJvaqY
- https://aclanthology.org/W17-7536.pdf

(The project may require ethics approval, depending on the selected dataset and evaluation method.)

Genre classification of songs, based on lyrics and metadata

The aim of this project is to build a system that identifies the genre of a song (e.g., pop, rock, jazz), using song lyrics and related metadata. Instead of classifying songs, the project can focus on classifying artists.
Some of the potential datasets or data sources that can be used are the following:
~~- Spotify API for a range of song metadata, including genres per artist and audio features.~~
- Song Interpretation Dataset for lyrics and metadata.
- Datasets containing information about artists and genres.

Prerequisites:
- Interest in Natural Language Processing and Machine Learning.
- Good programming skills (Python)
- Taking the module COM6115: Text Processing is highly recommended.

(The project may require ethics approval, depending on the selected dataset and evaluation method.)

Lyrics-based classification of songs

The aim of this project is to build a system that classifies a song using its lyrics and other lyrics-derived features or song metadata.
Based on their interests, students can choose from genre classification, artist classification, mood estimation, etc.

Some of the potential datasets or data sources that can be used are the following:
- Million Song Dataset
- Song Interpretation Dataset

Prerequisites:
- Interest in Natural Language Processing and Machine Learning.
- Good programming skills (Python)

Recommended reading:
- Lyrics Information Processing: Analysis, Generation, and Applications

The project may require ethics approval, depending on the selected evaluation method.

Lyrics-based song visualisation for music exploration

The aim of this project is to build a user interface for music exploration using song lyrics and lyrics-derived features, as well as song metadata.

Some of the potential datasets that can be used are the following:
- Million Song Dataset
- Song Interpretation Dataset

Recommended readings:
- Intelligent User Interfaces for Music Discovery
- LyricsRadar: A Lyrics Retrieval System Based on Latent Topics of Lyrics
- Lyrics Information Processing: Analysis, Generation, and Applications

The project may require ethics approval, as human participants may be involved in its evaluation.

A Web Application for a Student Marketplace

There are many different platforms enabling users to buy, sell or exchange clothes, furniture, books and other items. However, these platforms typically serve a wide geographic area and may not foster community-level connections.

This project aims to design and develop a web application that facilitates the exchange, sale, and purchase of various items (such as clothes, books, appliances, and other household goods) among students living in the same city and/or studying at the same University. The platform will serve as a localised marketplace but focused specifically on university or college students. Especially when moving to a new accommodation, students may face challenges with moving items from one place to another and may seek affordable ways to furnish their apartments, buy second-hand clothes, or find other used goods. The platform may not only provide access to affordable second-hand items available nearby, but also enhance community engagement and, potentially, help users meet other students, while encouraging a more sustainable lifestyle.

Here are some examples of similar applications:

1. https://www.brighton.ac.uk/news/2021/student-entrepreneurs-launch-online-marketplace
2. https://www.socialresponsibility.manchester.ac.uk/all-news/the-launch-of-hazaar-the-zero-waste-student-marketplace/

Comparison of different topic modelling approaches on song lyrics

Song lyrics often embed rich themes such as love, protest, identity, or storytelling, making them a valuable resource for exploring natural language processing methods. Topic modelling provides a way to uncover latent themes in large text corpora, yet traditional approaches (e.g. LDA) differ significantly from modern neural techniques (e.g. BERTopic, Top2Vec).

Various topic modelling approaches have been applied to song lyrics in previous research. Some examples include: analysis of the relationship between the musical and the lyrical content of metal music [1], understanding of how gender bias in lyrics varies thematically and across genres [2] and a music exploration web application based on thematic similarity of artists and songs [3].

This project proposes a systematic comparison of topic modelling methods on song lyrics, highlighting trade-offs between interpretability, coherence, and computational efficiency.

Main aims (can be adjusted):
1. Implement and compare classical (LDA – Latent Dirichlet Allocation, NMF – Non-negative Matrix Factorisation, PLSA – Probabilistic Latent Semantic Analysis) and neural/embedding-based (BERTopic, Top2Vec, CTM – Combined Topic Models) approaches.
2. Evaluate models using intrinsic metrics (e.g. coherence, diversity, perplexity) and/or extrinsic tests (e.g. genre classification, mood prediction).
3. Analyse how lyric characteristics (short texts, high repetition, figurative language) affect topic modelling performance.

Prerequisites:
- Interest in Natural Language Processing and Machine Learning.
- Good programming skills (Python)
- Taking the module COM3110/4115: Text Processing is highly recommended.

Recommended readings:
[1] Charting the Universe of Metal Music Lyrics and Analyzing Their Relation to Perceived Audio Hardness
[2] Beats of Bias: Analyzing Lyrics with Topic Modeling and Gender Bias Measurements
[3] Lyric Jumper: A Lyrics-based Music Exploratory Web Service by Modeling Lyrics Generative Process
[4] Latent Dirichlet Allocation
[5] BERTopic: Neural topic modeling with a class-based TF-IDF procedure
[6] Top2Vec: Distributed Representations of Topics
[7] A Topic Modeling Comparison Between LDA, NMF, Top2Vec, and BERTopic to Demystify Twitter Posts
[8] Topic Modeling Users' Interpretations of Songs to Inform
Subject Access in Music Digital Libraries
[9] TopicNet: Making Additive Regularisation for Topic Modelling Accessible

The project may require ethics approval, depending on the selected dataset and evaluation method (e.g. human evaluation of topic interpretability).

Lyrics Reconstruction and Synthetic Detection

Kim & Choi (2024) introduced LyCon, a method for reconstructing lyrics from bag-of-words datasets using large language models (LLMs). This approach addresses copyright restrictions that limit the sharing of lyrics datasets for academic research, while still preserving key attributes such as mood and genre.

Meanwhile, Labrak et al. (2024) explored the detection of synthetic lyrics across languages and genres, demonstrating that few-shot learning can reliably distinguish AI-generated lyrics, even when faced with unseen models and styles.

This project builds on and integrates these strands by:
- generating reconstructed lyrics using LyCon-style methods, and
- developing techniques to evaluate and detect synthetic lyrics (e.g. classification and/or clustering methods, as well as detection methods).

The project investigates both creative generation and authenticity detection, offering a rounded perspective on the role of AI in lyric production. The topic can be adapted to the specific interests and background of each student.

Prerequisites:
- Interest in Natural Language Processing and Machine Learning.
- Good programming skills (Python)
- Taking the module COM6115: Text Processing is highly recommended.

The project may require ethics approval, depending on the selected dataset and evaluation method.

Smart Gig Discovery: Location and Preference Based Event Recommendations in Sheffield

This project will design and develop a location aware web application that recommends live events to users in Sheffield. The system will allow users to set a search radius, select a preferred mode of transport such as walking, cycling, public transport or driving, and choose a category of gig. Based on these inputs, the application will generate a tailored list of events, incorporating geolocation data and travel time calculations to ensure recommendations are both relevant and accessible.

The focus of the dissertation is on Human Computer Interaction and Web Development, with an emphasis on usability, personalisation and user experience. The study will investigate how interactive design choices and location based technologies can support event discovery and encourage engagement with local cultural activities.

The project will include
- Collecting and organising local gig listings from open sources and event platforms
- Designing and implementing a recommendation system that accounts for user preferences and travel constraints
- Developing a web based interface that presents recommendations in a clear and engaging format
- Conducting user testing with Sheffield residents to evaluate usability, relevance and overall experience

The outcome of the dissertation will be a working prototype and an evaluation of its effectiveness in supporting user centered event discovery.